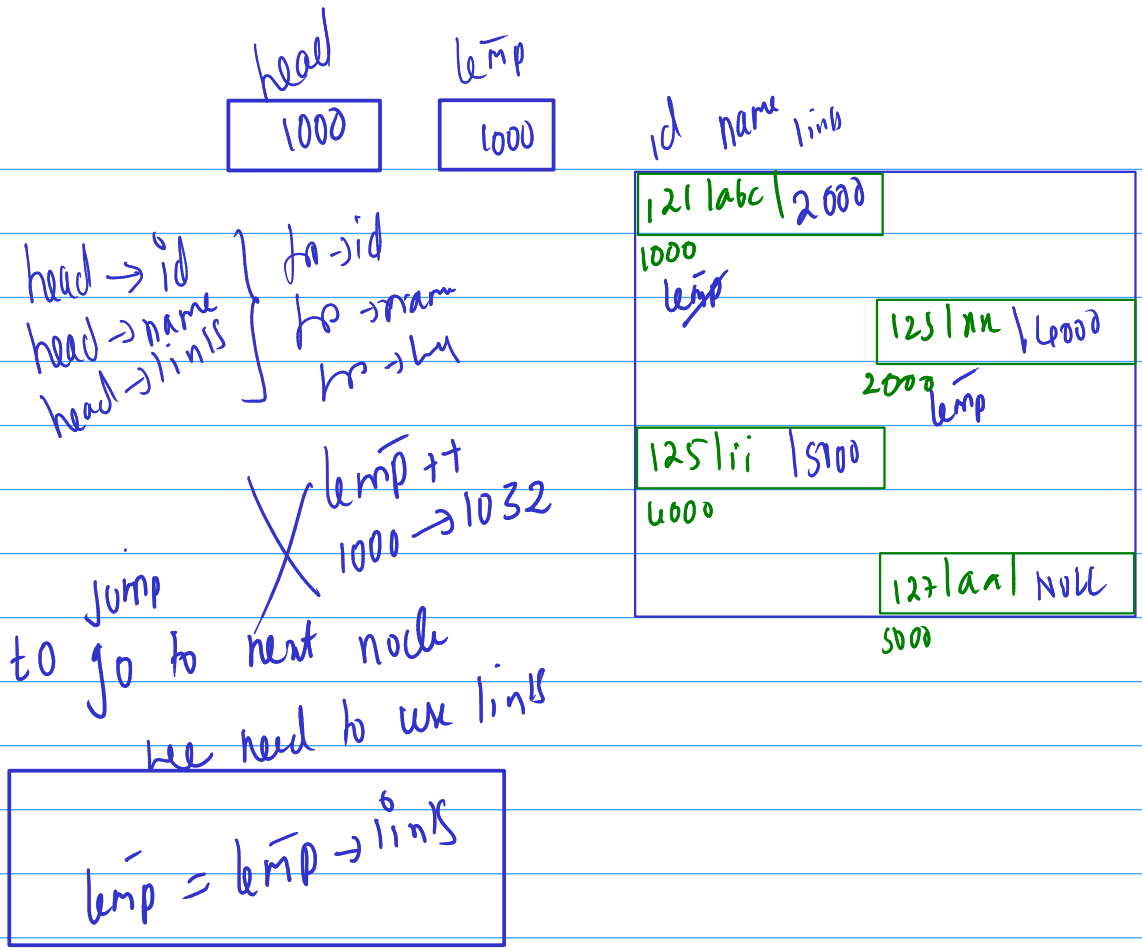
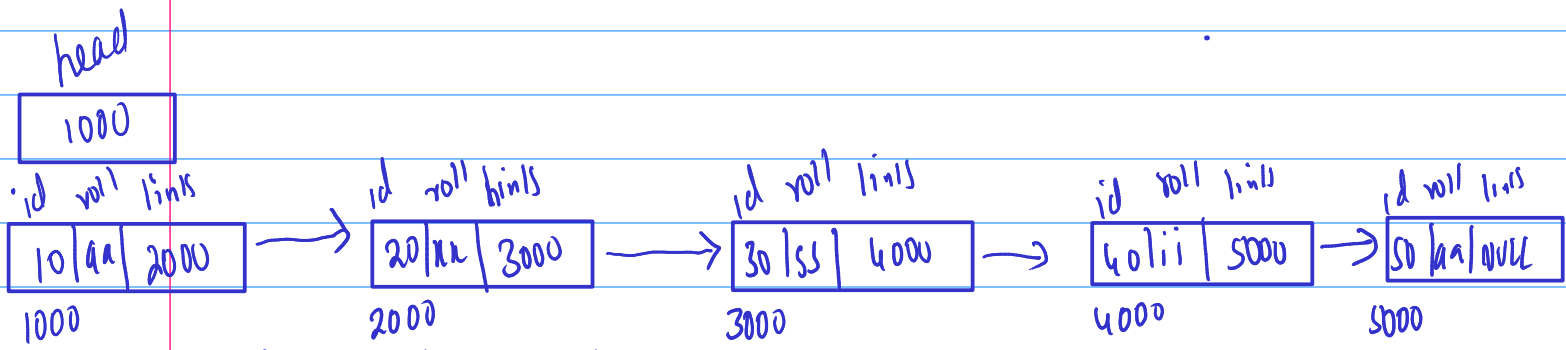
27/03



Single Link List : Singly link list is a collection of nodes, where every node is a combination of current node information and next node address.



Points to be noted :

1. in singly link list we can only traverse in forward direction
2. first node address should be storesd in a pointer called head
3. last node link should be NULL , to indicate last node of th list.

operations of Singly linked list

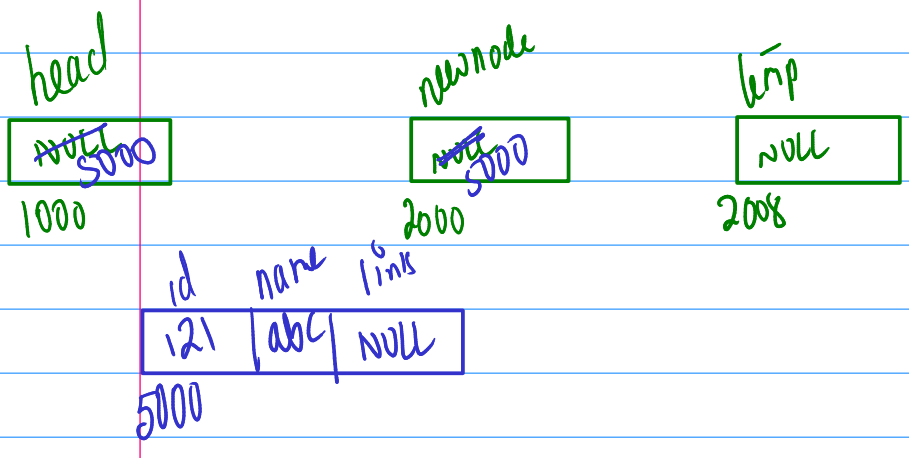
1. insert / Add
2. remov / delete
3. traverse
4. search

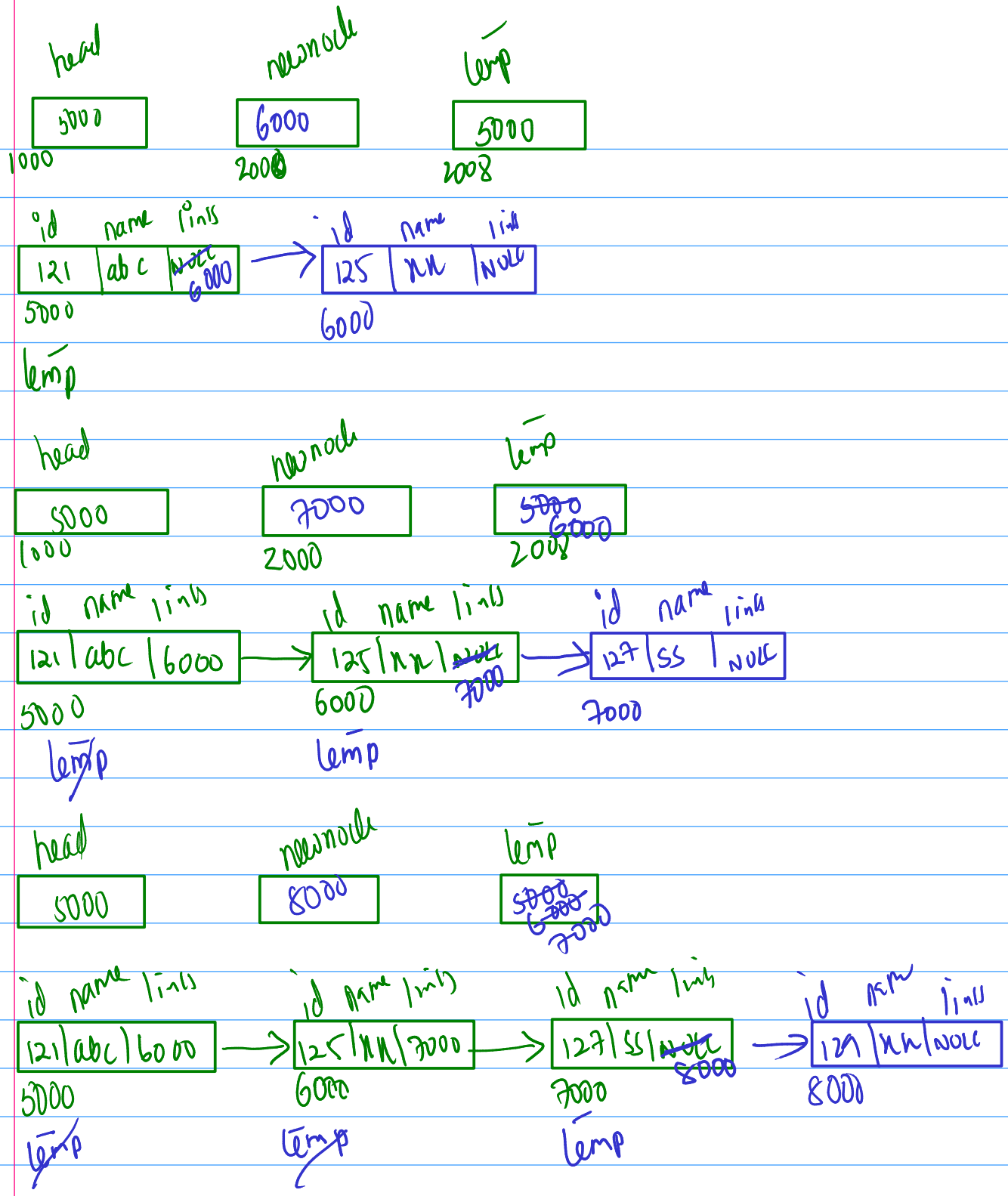
Addlast
Print
DelFirst
Save
Sync

```
Struct student
{
        int id;
        char name[20];
        struct Student *link;
};
```

Algorithm for Addlast

1. create a node
2. iniatialise the node
3. link to list
   a. if list is empty, assign newly created node address head
   b. traverse upto last node,
   assign newly created node address to last node link.
4. return firstnode address





creation of node :

```
newnode=calloc(1,sizeof(struct Student));
sf("%d",&newnode->id);
sf("%s",newnode->name);
// newnode->link=NULL; // must if malloc used
```

```
link to list
if(head==NULL)
        head=newnode;
else
{
        temp=head;
        while(temp->link!=NULL) // whilke(temp->link)
        {
                temp=temp->link;
        }
temp->link=newnode;
```

1. if list empty , print empty msg
2. else visit every node and print its information and move to next , until we reach end of the list

